

Bitcoin: A Peer-to-Peer Electronic Cash System

A technical overview of the Bitcoin protocol, prepared as course material.

Abstract

Bitcoin is a decentralised digital currency that allows online payments to be sent directly from one party to another without going through a financial institution. The system relies on a distributed timestamp server that produces computational proof of the chronological order of transactions.

Chapter 1 · Transactions and the UTXO Model

The UTXO Model

Bitcoin does not track account balances directly. Instead, the state of the ledger is a set of Unspent Transaction Outputs (UTXOs). A transaction consumes one or more existing UTXOs as inputs and creates one or more new UTXOs as outputs.

Transaction Structure

Each transaction references its inputs by the hash of the previous transaction and the index of the output being spent. The sum of the input values must be greater than or equal to the sum of the output values; the difference is collected by the miner as a fee.

Field	Size	Description
version	4 bytes	Transaction format version
tx_in count	varint	Number of transaction inputs
tx_in	variable	List of transaction inputs
tx_out count	varint	Number of transaction outputs

Chapter 2 · Proof of Work and Consensus

Proof of Work

To implement a distributed timestamp server, Bitcoin uses a proof-of-work system. Miners scan for a value that, when hashed with SHA-256, produces a hash beginning with a required number of zero bits.

The difficulty condition can be expressed as:

$H(\text{block_header} \parallel \text{nonce}) < \text{target}$

Chain Selection

Nodes always consider the longest valid chain to be the correct one and will keep working on extending it. If two miners broadcast different versions of the next block simultaneously, nodes work on the first one they received, but save the other branch in case it becomes longer.

Chapter 3 · Merkle Trees and Simplified Verification

Merkle Trees

Transactions in a block are organised into a Merkle tree, with only the root included in the block's hash. This allows a transaction to be verified without downloading the entire block.

Simplified Payment Verification

A user can verify that a transaction was accepted by the network by keeping a copy of the block headers of the longest proof-of-work chain and obtaining the Merkle branch linking the transaction to the block it is timestamped in.

Summary

The combination of proof-of-work chain selection, the UTXO model, and Merkle-tree-based verification allows Bitcoin to operate without a trusted central authority.